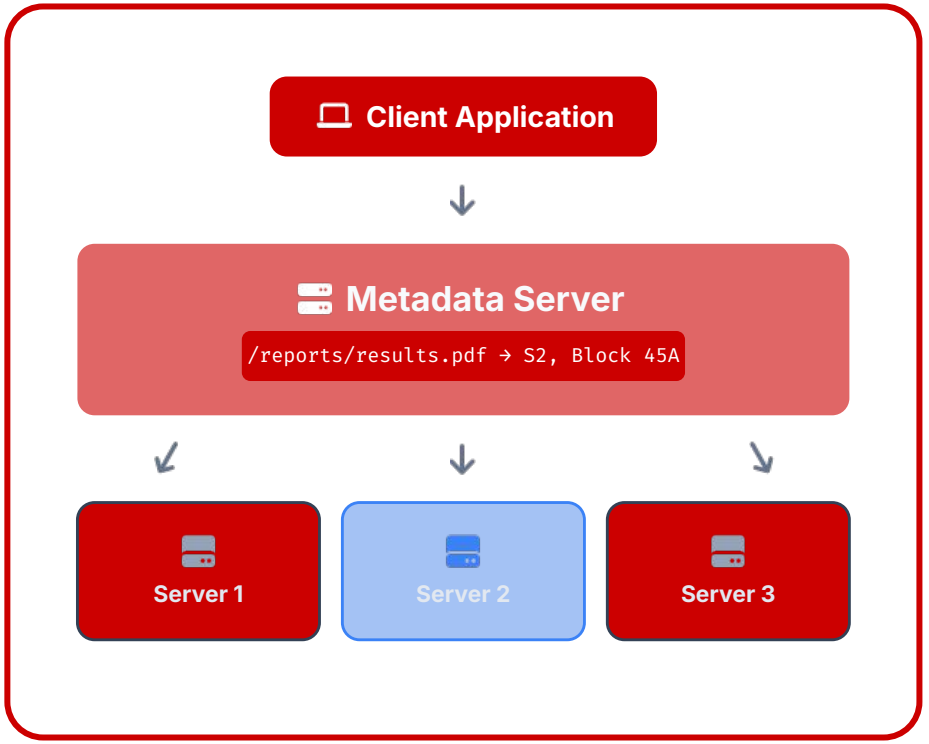
SYSTEM ARCHITECTURE

Location Transparency

Decoupling logical file paths from physical storage locations.

I implement a centralized **Metadata Server** to handle the naming scheme. It maintains an internal database mapping logical paths (e.g., `/shared/reports/results.pdf`) to physical locations (e.g., Server 2). This abstraction layer ensures clients interact only with logical names, completely decoupling access from physical infrastructure.



EXECUTION TRACE

Read Request Walkthrough

SOLUTION EXPLANATION

The read process is split to avoid bottlenecks.

1) The client asks the Metadata Server to resolve the logical path.

2) The server checks authorization and returns the physical server's IP along with a secure cryptographic token.

3) The client connects directly to the storage server, bypassing the Metadata Server for heavy data streaming.

1

Name Resolution

Client requests logical file path.

2

Authorization

Metadata server issues cryptographic token.

3

Data Streaming

Client downloads directly from storage.

Handling Stale Location Data

SOLUTION EXPLANATION

When a file moves, the Metadata Server updates its mapping transactionally. If a client attempts to use a cached, outdated location, the old storage server proactively rejects the request with a "Stale Handle" error. This triggers the client to automatically flush its local cache and fetch the updated routing from the Metadata Server.

1. **Client queries old server from cache.**
2. **Server returns "Stale Handle" error.**
3. **Client automatically flushes cache.**
4. **Client requests new path from metadata.**

Failures & Design Trade-offs

SOLUTION EXPLANATION

Location transparency is not fault transparency: if a storage node crashes, data is inaccessible without explicit replication. Furthermore, the centralized Metadata Server represents a Single Point of Failure. Finally, security must not be compromised: the Metadata Server verifies access rights before revealing locations, and storage nodes enforce them using crypto-tokens.

Storage Node Failure

Location transparency cannot prevent data loss.

Metadata Server Failure

Global system halt without clustered metadata nodes.

Security vs Scalability

Centralized authorization creates potential performance bottlenecks.